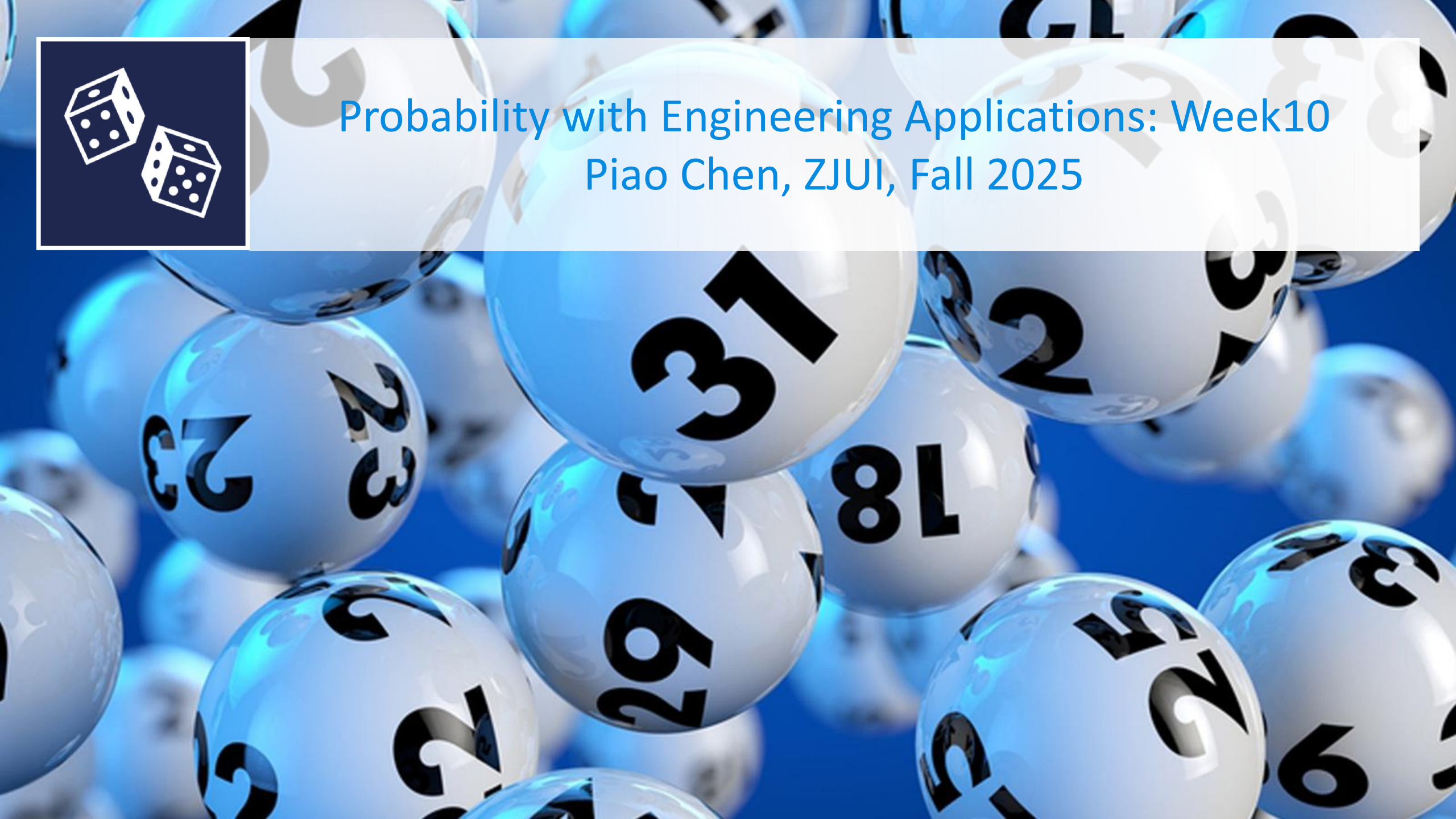




Probability with Engineering Applications: Week10

Piao Chen, ZJUI, Fall 2025



Programme

- Function of two random variables
- Covariance and correlation
- Bivariate normal

Distribution of function of random variables

General Transformations

Let us now consider general mappings:

$$\begin{pmatrix} W \\ Z \end{pmatrix} = g \left[\begin{pmatrix} X \\ Y \end{pmatrix} \right]$$

The function g maps points from the u - v plane to the α - β plane

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} g_1(u, v) \\ g_2(u, v) \end{pmatrix}$$

Def. The **Jacobian** of g is given by

$$J = J(u, v) = \begin{pmatrix} \frac{\partial g_1(u, v)}{\partial u} & \frac{\partial g_1(u, v)}{\partial v} \\ \frac{\partial g_2(u, v)}{\partial u} & \frac{\partial g_2(u, v)}{\partial v} \end{pmatrix}$$

General Transformations

Prop. Suppose $\begin{pmatrix} W \\ Z \end{pmatrix} = g \left[\begin{pmatrix} X \\ Y \end{pmatrix} \right]$ where the pair of random variables (X, Y) have joint pdf $f_{X,Y}$, and g is a **one-to-one mapping** from the support of $f_{X,Y}$ to $\mathbb{R}^2$. Suppose the Jacobian $\mathbf{J}$ of g **exists**, is **continuous**, and has **nonzero determinant everywhere**. Then the pair of random variables (W, Z) has joint pdf given by

$$f_{W,Z}(\alpha, \beta) = \frac{1}{|\det(\mathbf{J})|} f_{X,Y} \left(g^{-1} \left[\begin{pmatrix} \alpha \\ \beta \end{pmatrix} \right] \right)$$

for points (α, β) in the support of $f_{W,Z}$.

Example

Let X and Y have joint pdf

$$f_{X,Y}(u, v) = \begin{cases} u + v, & (u, v) \in [0, 1]^2 \\ 0, & \text{otherwise} \end{cases}$$

Let $W = X^2$ and $Z = X(1 + Y)$. Find $f_{W,Z}$.



Example

Linearity of expectations

Linearity of expectations

Theorem:

For every two random variables X and Y it holds that

$$E[rX + sY + t] = rE[X] + sE[Y] + t$$

for all values r, s and t .



Correlation

Correlation

Sometimes (dependent) RVs X and Y are also correlated.

Examples:

For a randomly selected student:

X = hours spent on the course

Y = grade on the midterm

If X is high the probability that Y is high increases

If X is low the probability that Y is low increases

This is called positive correlation



Correlation

Sometimes (dependent) RVs X and Y are also correlated.

Examples:

For a randomly selected student:

X = time spent on exercises

Y = the body weight

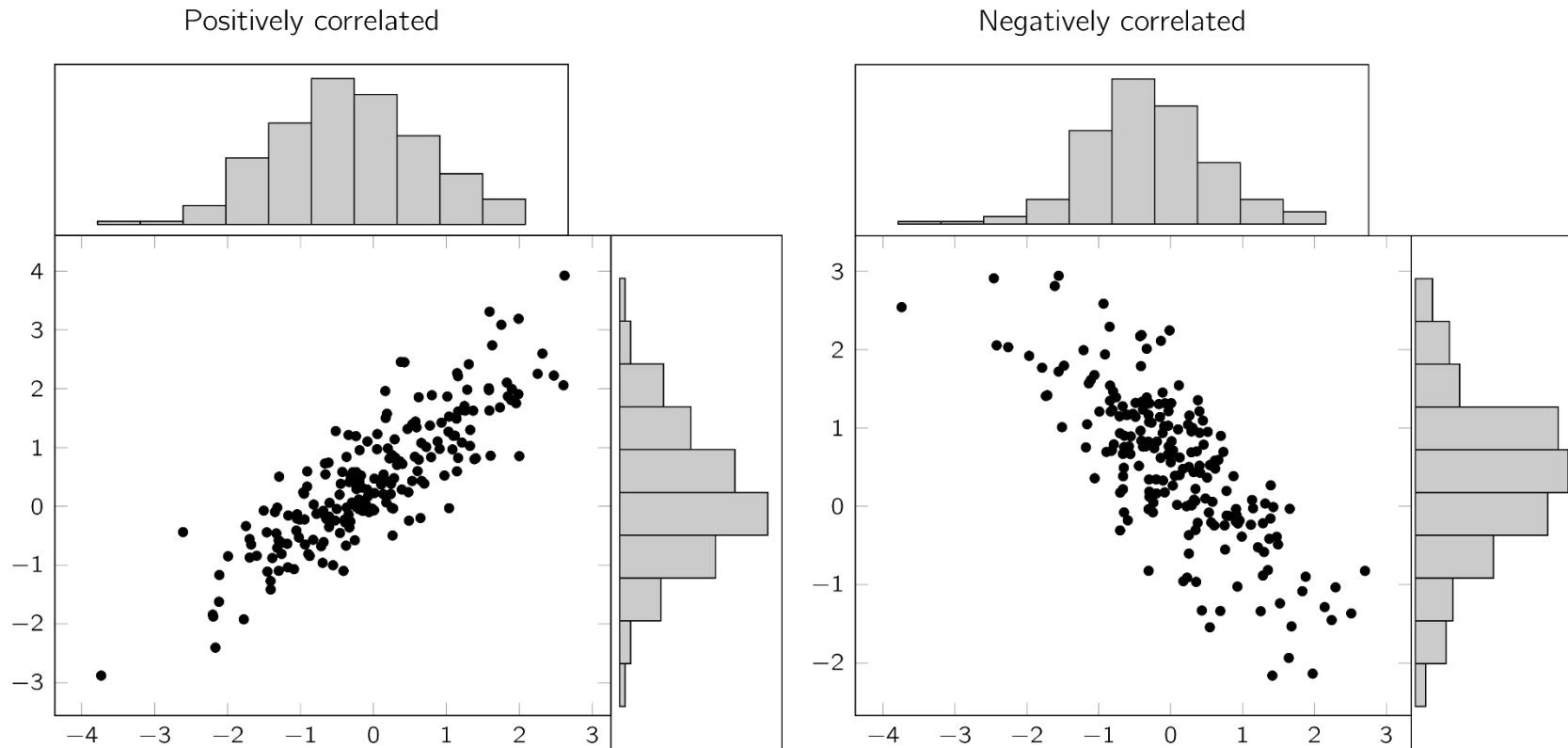
If X is high the probability that Y is low increases

If X is low the probability that Y is high increases

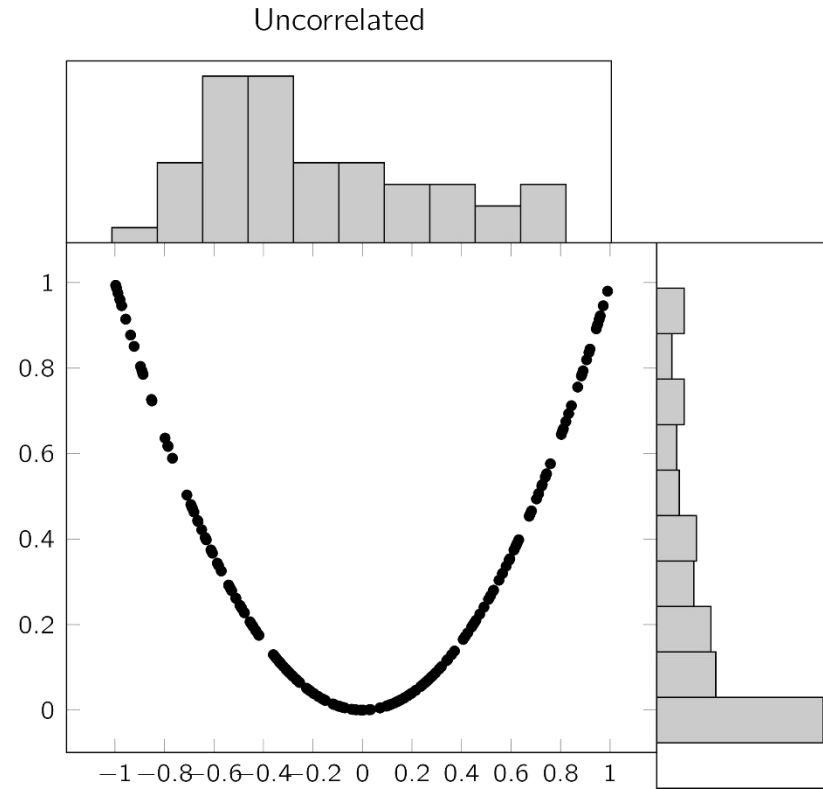
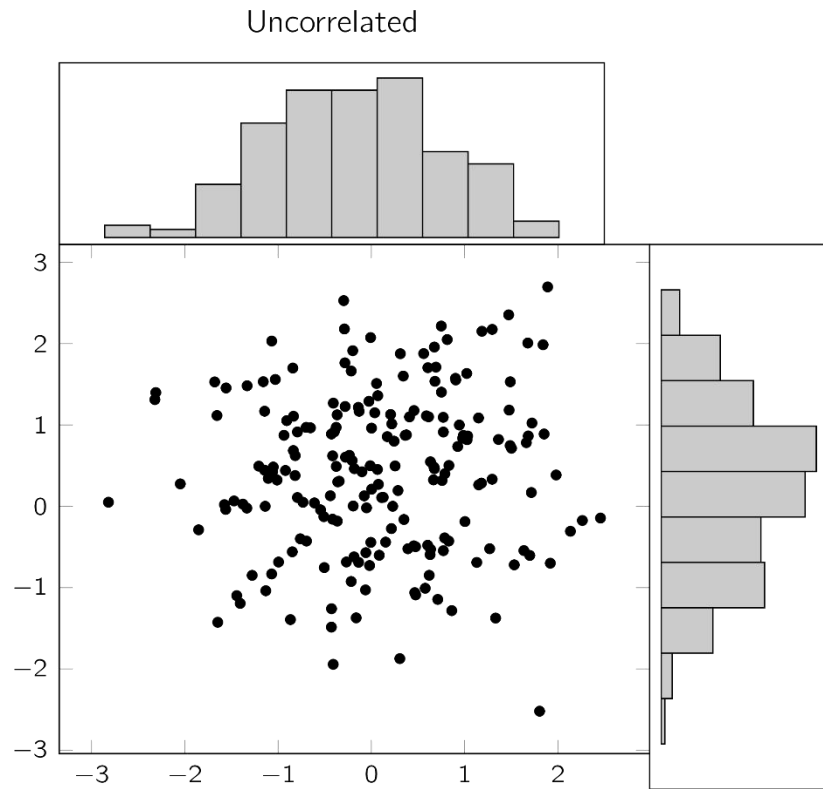
This is called negative correlation



Correlation in a scatterplot



Correlation in a scatterplot



Covariance

Covariance

Definition:

Let X and Y be two RVs. The covariance between X and Y is

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])].$$

If $\text{Cov}(X, Y) > 0$, then X and Y are positively correlated.

If $\text{Cov}(X, Y) < 0$, then X and Y are negatively correlated.

If $\text{Cov}(X, Y) = 0$, then X and Y are uncorrelated.

A useful way to compute the covariance is

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y].$$



Practice now

Consider the two discrete random variables X and Y for which you can see the joint probability mass function in the following table:

b	a			$P(Y = b)$
	0	1	2	
-1	1/6	1/6	1/6	1/2
1	0	1/2	0	1/2
$P(X = a)$	1/6	2/3	1/6	1

- Determine $E[XY]$.
- Note that X and Y are dependent. Show that X and Y are uncorrelated.



Independent vs uncorrelated

Covariance

If two RVs X and Y are independent, then they are also uncorrelated. However, uncorrelated RVs may be dependent.

Theorem:

Let X and Y be two RVs, then always

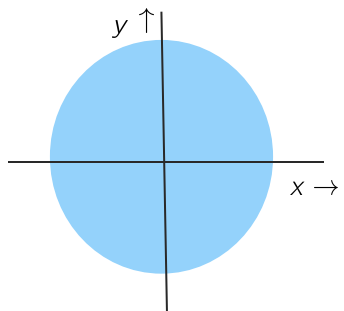
$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y).$$

If X and Y are uncorrelated, then

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$



Coins in Ponds

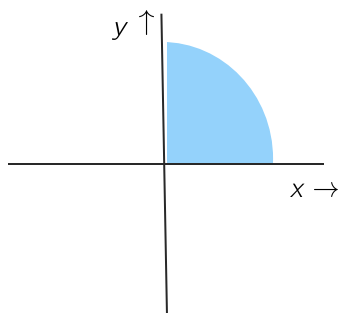


$$f_X(x) = \frac{\sqrt{4 - x^2}}{2\pi} \quad \text{if } -2 \leq x \leq 2$$

$$f_Y(y) = \frac{\sqrt{4 - y^2}}{2\pi} \quad \text{if } -2 \leq y \leq 2$$

$$f(x, y) = \begin{cases} \frac{1}{4\pi} & \text{if } \sqrt{x^2 + y^2} \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Coins in Ponds



$$f_X(x) = \frac{\sqrt{4-x^2}}{\pi} \quad \text{if } 0 \leq x \leq 2$$

$$f_Y(y) = \frac{\sqrt{4-y^2}}{\pi} \quad \text{if } 0 \leq y \leq 2$$

$$f(x, y) = \begin{cases} \frac{1}{\pi} & \text{if } \sqrt{x^2 + y^2} \leq 2, x \geq 0, y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Change of units

Change of units

Theorem:

Let X and Y be two RVs. Then

$$\text{Cov}(rX + s, tY + u) = rt\text{Cov}(X, Y)$$

for all values r, s, t and u .



Correlation coefficient

Correlation coefficient

Definition:

Let X and Y be two RVs. The correlation coefficient is

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}}$$

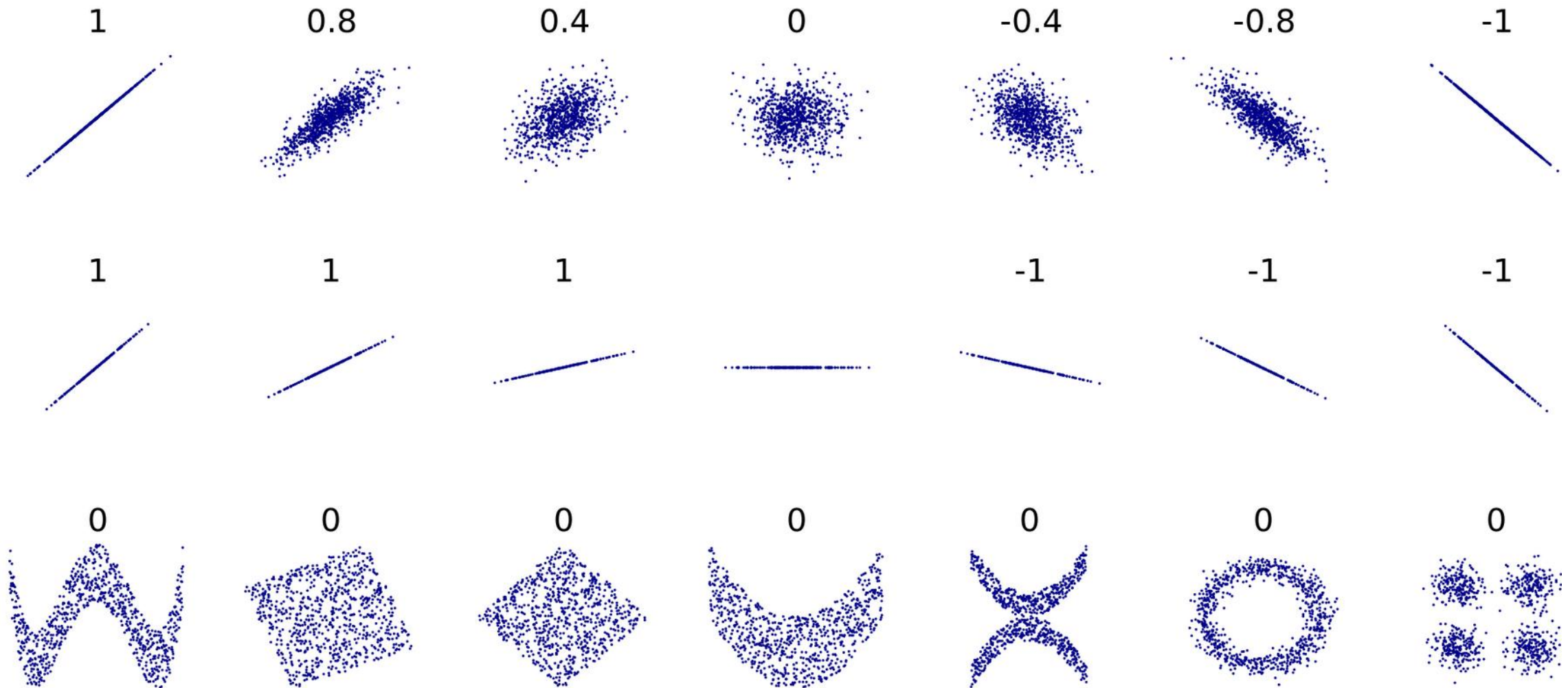
if $\text{Var}(X) > 0$ and $\text{Var}(Y) > 0$. Otherwise, $\rho(X, Y) = 0$.

Properties:

- $-1 \leq \rho(X, Y) \leq 1$
- $\rho(rX + s, tY + u) = \begin{cases} \rho(X, Y) & \text{if } rt > 0, \\ -\rho(X, Y) & \text{if } rt < 0. \end{cases}$



Correlation coefficient



Linearity

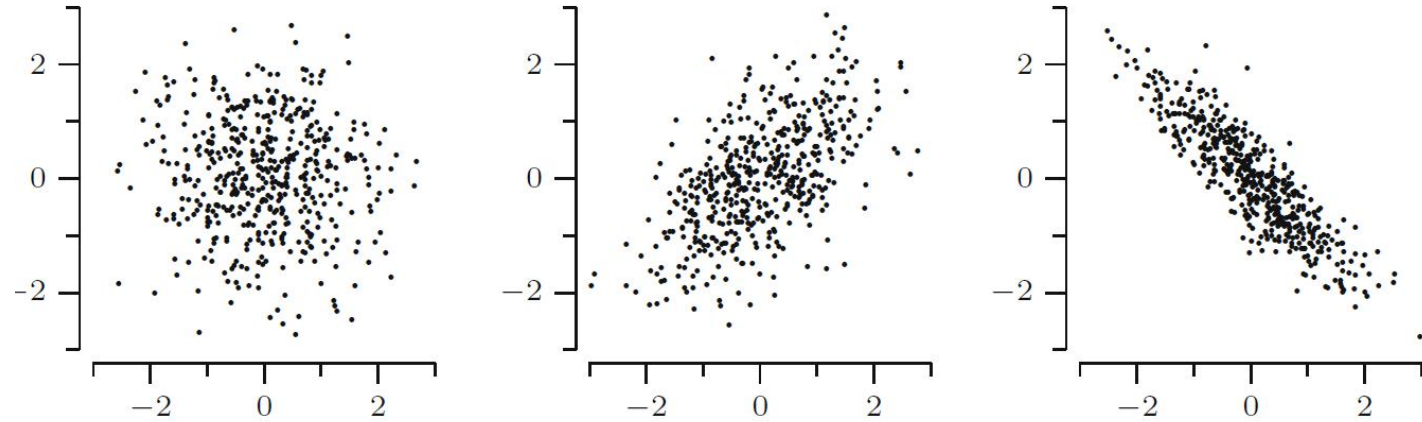
Consider the following statements.

1. For any two RVs X and Y , $E[X + Y] = E[X] + E[Y]$
2. For independent RVs X and Y , $\text{Var}(X - Y) = \text{Var}(X) - \text{Var}(Y)$

Which of these statements is/are true?

- A.** None
- B.** Only 1
- C.** Only 2
- D.** Both

Practice now



In the figure above three plots are displayed. For each plot we carried out a simulation in which we generated 500 realizations of a pair of random variables (X, Y) . We have chosen three different joint distributions of X and Y .

- Indicate for each plot whether it corresponds to random variables X and Y that are positively correlated, negatively correlated, or uncorrelated.
- In which plot is $|\rho(X, Y)|$ maximal?



Bivariate Normal Distribution

Revisit the Normal Distribution

Recall the definition of a Gaussian random variable $X \sim N(\mu, \sigma^2)$

$$f_X(u) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(u - \mu)^2}{2\sigma^2} \right], \quad -\infty < u < \infty$$

2 parameters:

- μ = mean
- σ^2 = variance

Bivariate Normal Distribution

Def. Random variables X and Y are said to be **jointly Gaussian** if **every** linear combination $aX + bY$ is a Gaussian random variable.

For the purpose of the above definition, a constant is considered to be a Gaussian random variable with zero variance

The **bivariate normal / Gaussian** probability density function is

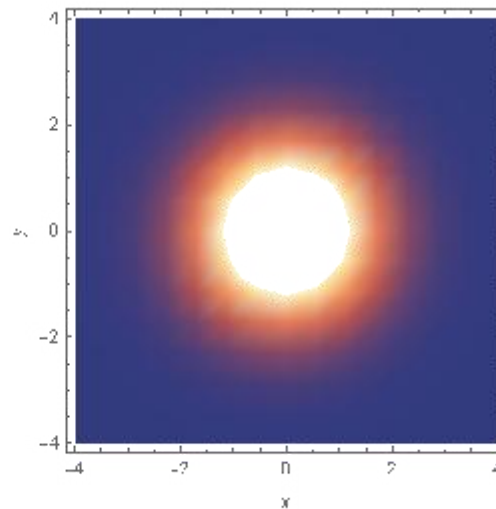
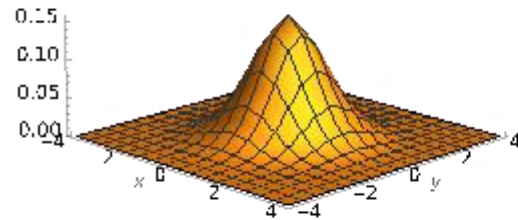
$$f_{X,Y}(u, v) = C \exp \left[- \frac{\left(\frac{u - \mu_X}{\sigma_X} \right)^2 + \left(\frac{v - \mu_Y}{\sigma_Y} \right)^2 - 2\rho \left(\frac{u - \mu_X}{\sigma_X} \right) \left(\frac{v - \mu_Y}{\sigma_Y} \right)}{2(1 - \rho^2)} \right]$$
$$C = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1 - \rho^2}}$$

Bivariate Normal PDF Parameters

The bivariate normal pdf has 5 parameters:

- μ_X and μ_Y (center location)
- σ_X and σ_Y (scale)
- ρ (correlation)

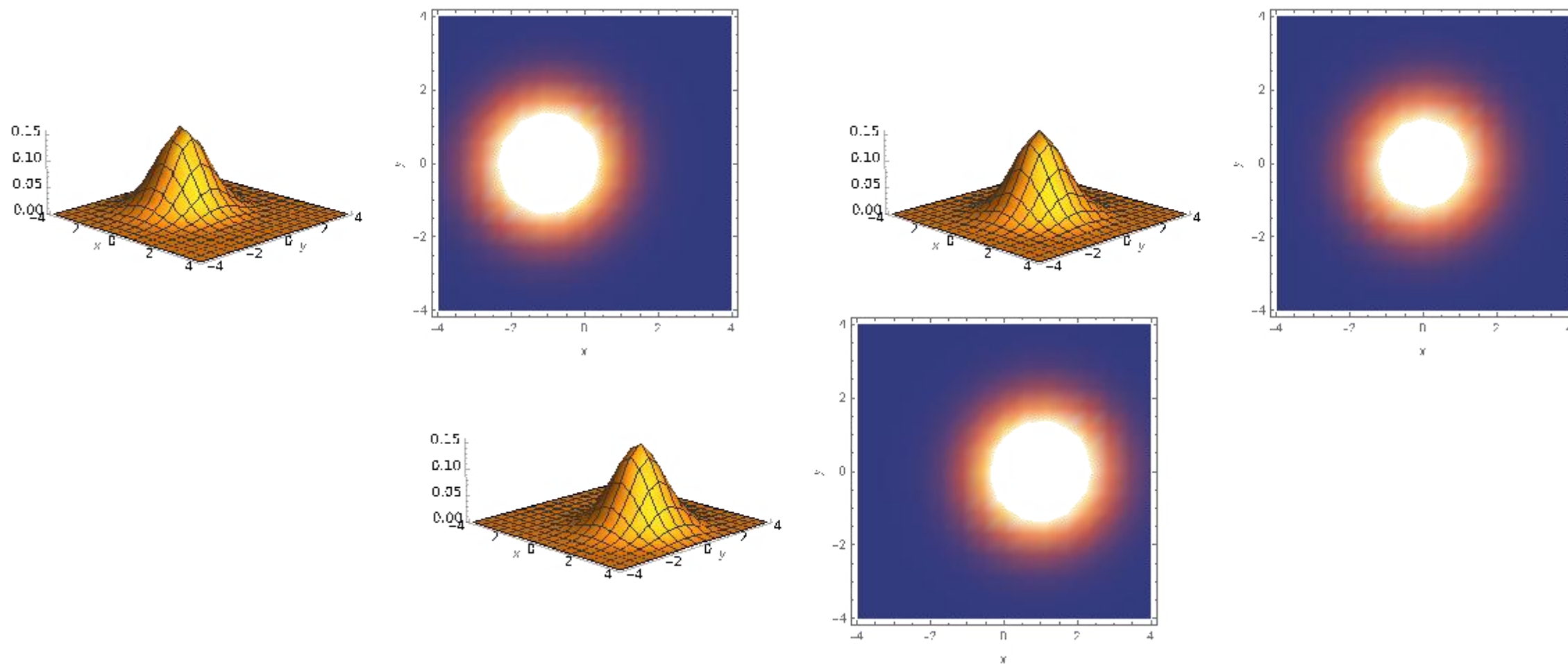
Consider: $\mu_X = \mu_Y = 0, \sigma_X = \sigma_Y = 1, \rho = 0$



Visualization of the Five Parameters

Consider: $\mu_Y = 0, \sigma_X = \sigma_Y = 1, \rho = 0$

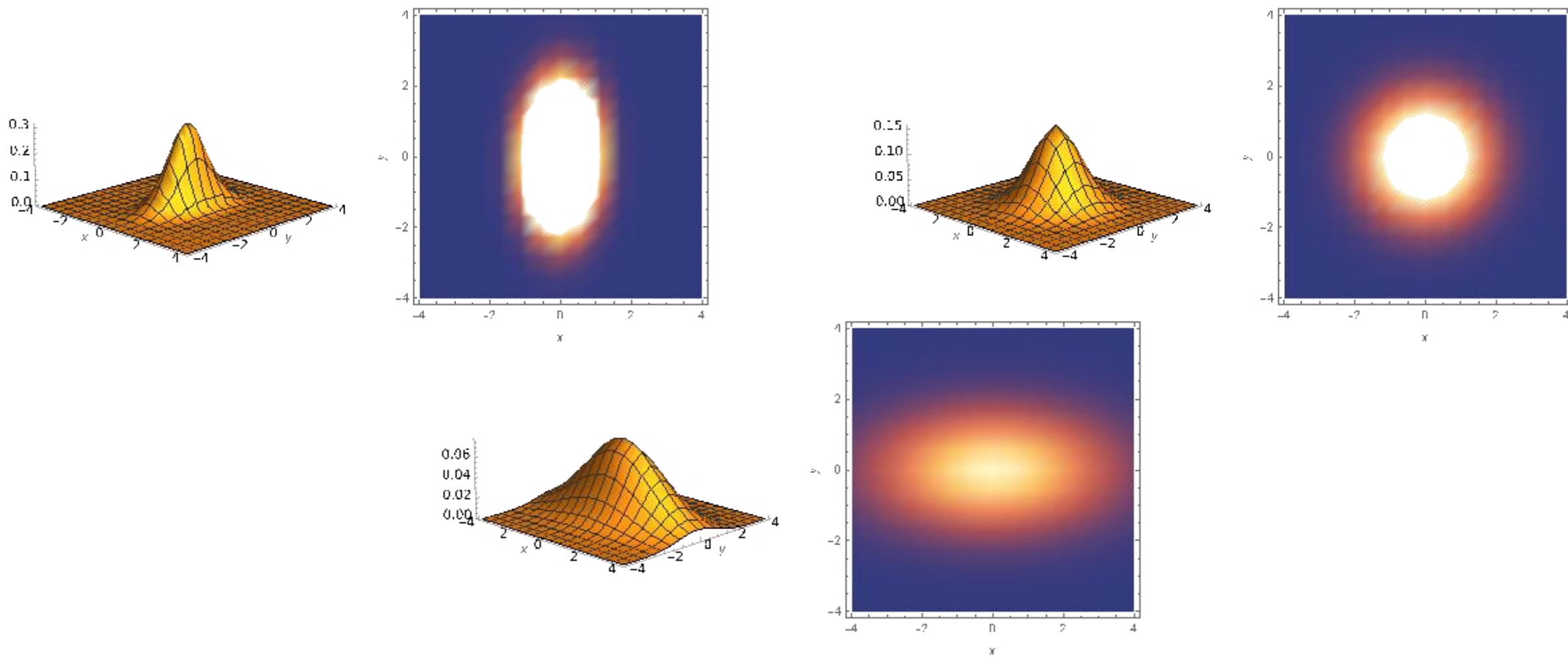
μ_X is changing: -1, 0, 1



Visualization of the Five Parameters

Consider: $\mu_X = \mu_Y = 0$, $\sigma_Y = 1$, $\rho = 0$

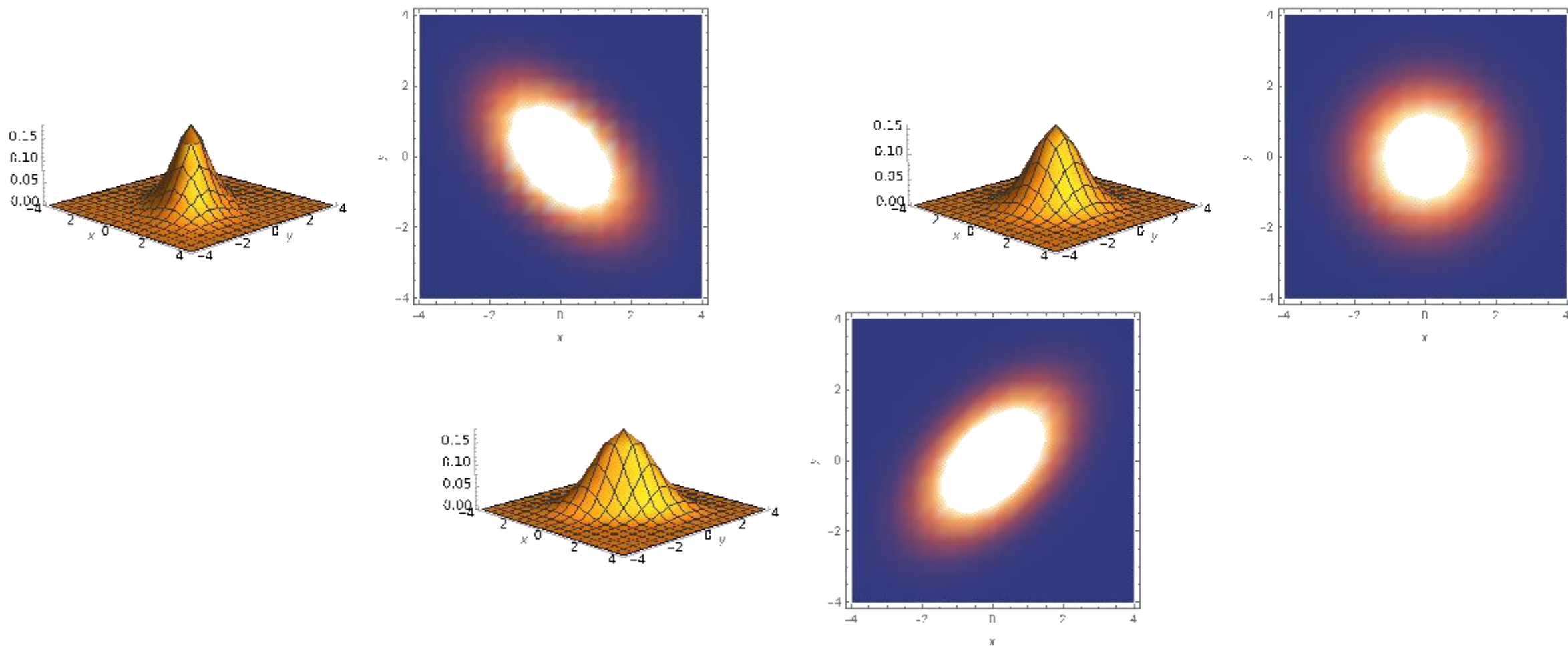
σ_X is changing: 0.5, 1, 1.5



Visualization of the Five Parameters

Consider: $\mu_X = \mu_Y = 0$, $\sigma_X = \sigma_Y = 1$

ρ is changing: -0.5, 0, 0.5



How to Determine if a PDF is Bivariate Normal?

A function is a bivariate normal pdf if can be written in the following form:

$$f_{X,Y}(u, v) = C \exp [-g(u, v)]$$

where $g(u, v) = au^2 + buv + cv^2 + du + ev + f$ and $a > 0, c > 0$, and $b^2 - 4ac < 0$ (see Hajek notes)

This can also be written as

$$f_{X,Y}(u, v) = C_2 \exp [-g_2(u, v)]$$

where $g_2(u, v) = au^2 + buv + cv^2 + du + ev$ and $C_2 = C \exp (-f)$

$\Rightarrow$ the 5 parameters $\{a, b, c, d, e\}$ uniquely determine the bivariate normal pdf

Properties of the Bivariate Normal Distribution

Zero Mean, Unit Variance Case

Suppose the joint pdf the random variables X and Y is bivariate normal. What are the implications?

Without loss of generality, we can consider the case where

$$- \mu_X = \mu_Y = 0$$

$$- \sigma_X^2 = \sigma_Y^2 = 1$$

In this case the bivariate normal pdf is given by

$$\begin{aligned} f_{X,Y}(u, v) &= \frac{1}{2\pi\sqrt{1-\rho^2}} \exp \left[-\frac{u^2 + v^2 - 2\rho uv}{2(1-\rho^2)} \right] \\ &= \left[\frac{1}{\sqrt{2\pi}} \exp \left(-\frac{u^2}{2} \right) \right] \left[\frac{1}{\sqrt{2\pi(1-\rho^2)}} \exp \left(-\frac{(v - \rho u)^2}{2(1-\rho^2)} \right) \right] \end{aligned}$$

Zero Mean, Unit Variance Case

- Note that $f_{X,Y}(u, v) = \left[\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) \right] \left[\frac{1}{\sqrt{2\pi(1-\rho^2)}} \exp\left(-\frac{(v-\rho u)^2}{2(1-\rho^2)}\right) \right]$
 $= g(u) h(u, v)$
- The function $g(u)$ is the pdf of a standard Gaussian, i.e., $N(0, 1)$
- For u fixed, the function $h(u, v)$ is the $N(\rho u, 1 - \rho^2)$ pdf
- Thus, we have found $f_{X,Y}(u, v) = f_X(u) f_{Y|X}(v|u)$
- We could also have factored the bivariate normal pdf as
$$f_{X,Y}(u, v) = g(v) h(v, u) = f_Y(v) f_{X|Y}(u|v)$$
$$[N(0, 1) N(\rho v, 1 - \rho^2)]$$

Important Properties of the Bivariate Normal Distribution

Prop. Suppose X and Y have the bivariate normal pdf, then:

1) $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$

Proof. Direct consequence of the factorization given on the previous slide.

Important Properties of the Bivariate Normal Distribution

Prop. Suppose X and Y have the bivariate normal pdf, then:

2) $aX + bY$ is normal for any $a, b \in \mathbb{R}$

Proof. Follows from the fact that linear transformations of the pair of random variables (X, Y) will also have the bivariate normal pdf (see Hajek notes).

Implications: let
$$A \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} aX + bY \\ cX + dY \end{pmatrix} = \begin{pmatrix} X' \\ Y' \end{pmatrix}$$

- By result 1), X' is normal
- Then, 2) implies that X and Y are jointly normal

Important Properties of the Bivariate Normal Distribution

Prop. Suppose X and Y have the bivariate normal pdf, then:

3) ρ is the correlation coefficient between X and Y

Important Properties of the Bivariate Normal Distribution

Prop. Suppose X and Y have the bivariate normal pdf, then:

4) X and Y are independent if and only if $\rho = 0$

Important Properties of the Bivariate Normal Distribution

Prop. Suppose X and Y have the bivariate normal pdf, then:

5) The conditional distribution of Y given $X = u$ is $N(E[Y|X = u], \sigma_e^2)$

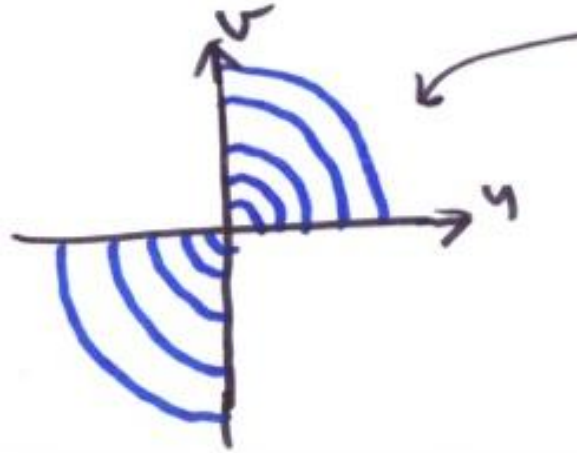
Bivariate normal = marginal normals?

What is an example where X and Y are each normally distributed but **not bivariate normal**?

Consider $X \sim N(0, 1)$ and $Y \sim N(0, 1)$. Suppose the joint density of X and Y is given by

$$f_{X,Y}(u, v) = \begin{cases} 2f_X(u)f_Y(v), & u \geq 0 \text{ and } v \geq 0, \text{ or } u < 0 \text{ and } v < 0 \\ 0, & \text{otherwise} \end{cases}$$

What isn't bivariate normal?



The contours indicate
equal probability density
in $f_{X,Y}(u, v)$

X and Y are **marginally** normal but the joint pdf $f_{X,Y}$ is not bivariate normal, i.e., X and Y are not **bivariate** normal

Multivariate Gaussian Distribution

It is not difficult to generalize the Gaussian distribution to n variables

Def. A **random vector** $\mathbf{X}$ with dimension n is an ordered collection of n random variables

$$\mathbf{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix}$$

Def. The expectation of a random vector is the vector of the expectations:

$$E[\mathbf{X}] = \begin{pmatrix} E[X_1] \\ \vdots \\ E[X_n] \end{pmatrix}$$

Multivariate Gaussian Distribution

Def. The covariance of random vectors $\mathbf{X}$ and $\mathbf{Y}$ with dimensions n and m , respectively, is the $n \times m$ matrix Σ where

$$\Sigma[i, j] = \text{Cov}(X_i, Y_j), \quad 1 \leq i \leq n, \quad 1 \leq j \leq m$$

Def. A random vector $\mathbf{X}$ is said to have the **n dimensional Gaussian density** with mean vector $\mu = [\mu_1, \dots, \mu_n]^T$ and $n \times n$ covariance matrix Σ if the joint pdf is given by

$$f_{\mathbf{X}}(\mathbf{u}) = \frac{1}{(2\pi)^{\frac{n}{2}} \sqrt{\det \Sigma}} \exp \left[-\frac{(\mathbf{u} - \mu)^T \Sigma^{-1} (\mathbf{u} - \mu)}{2} \right]$$



See you next week